VTTSI-Chat: Agentic GeoAI for Real-Time Intersection Safety Monitoring with Natural Language Explanation [Demo]

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Abstract

We present **VTTSI-Chat**, a cloud-native agentic GeoAI system that pairs the Virginia Tech Transportation Safety Index (VTTSI)—a hybrid Real-Time Safety Index (RT–SI) and CRITIC-weighted MCDM pipeline—with a **SafetyChat** conversational module powered by a tool-augmented GPT-4o. VTTSI fuses seven years of VDOT crash records via Empirical Bayes stabilization with live J2735 connected-vehicle telemetry stored in a PostGIS spatial database, producing interpretable 0–100 safety scores every 15 minutes across 18 instrumented intersections in Falls Church, VA. SafetyChat exposes six typed API tools to the LLM, grounding all natural language responses in live geospatial data and enabling TMC operators, planners, and emergency responders to query causal safety narratives through ordinary conversation. A live deployment at https://cs6604-trafficsafety-180117512369.europe-west1.run.app demonstrates four end-to-end use cases and is available for interactive exploration at the conference.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; **Intelligent agents**; • **Information systems** → *Decision support systems*; • **Human-centered computing** → *Interactive systems and tools.*

Keywords

agentic GeoAI, traffic safety, spatio-temporal safety index, natural language interface, MCDM, Empirical Bayes, connected vehicles, PostGIS, tool-augmented LLM

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1 Introduction

Geospatial intersection safety data are increasingly available through connected-vehicle infrastructure, yet a critical gap remains between numerical safety scores and actionable insight for operators. TMC practitioners must interpret multi-dimensional spatio-temporal indices under time pressure, yet existing dashboards present data without explaining causality or suggesting actions. Tool-augmented LLMs can reason over structured numerical data to produce grounded explanations [10], but prior LLM+traffic work targets signal control [5] or AV motion planning [6, 13]—not safety-index explanation for infrastructure operators.

VTTSI-Chat closes this gap by pairing the VTTSI hybrid scoring pipeline with **SafetyChat**: a GPT-4o module that issues typed function calls to live VTTSI endpoints and returns causal safety narratives via natural language dialogue. The system operates as an *agentic GeoAI* interface: the LLM agent queries live PostGIS-backed safety data, reasons over spatio-temporal patterns across 18 geospatially registered intersections, and synthesizes evidence-grounded narratives for diverse operator roles. VTTSI-Chat is the first system to combine a hybrid EB-stabilized crash index with real-time MCDM scoring *and* a natural language interface for spatial safety data.

2 VTTSI System Overview

The VTTSI backend (FastAPI on Google Cloud Run) computes two complementary safety indices every 15 minutes. Six real-time tables stream from the VTTI Trino connected-vehicle system into Cloud SQL PostGIS: `bsm` (J2735 Basic Safety Messages: position, speed, braking), `psm` (Personal Safety Messages from VRUs), `vehicle-count`, `vru-count`, `speed-distribution`, and `safety-event` (red-light running, intersection conflicts, near-miss detections). All streams are synchronized into 15-minute bins; a separate VDOT crash dataset (2017–2024) is spatially joined to intersections at import time via a PostGIS `ST_Within` predicate using the VDOT LRS centerline network.

2.1 Real-Time Safety Index (RT–SI)

RT–SI anchors intersection risk to a severity-weighted Empirical Bayes (EB) baseline derived from 2017–2024 VDOT crash records. The historical crash rate is:

$$r_i = \frac{\sum_s \ w_s \, C_{i,s}}{E_i}, \quad w_{\text{Fatal}} = 10,\ w_{\text{Inj}} = 3,\ w_{\text{PDO}} = 1, \qquad (1)$$

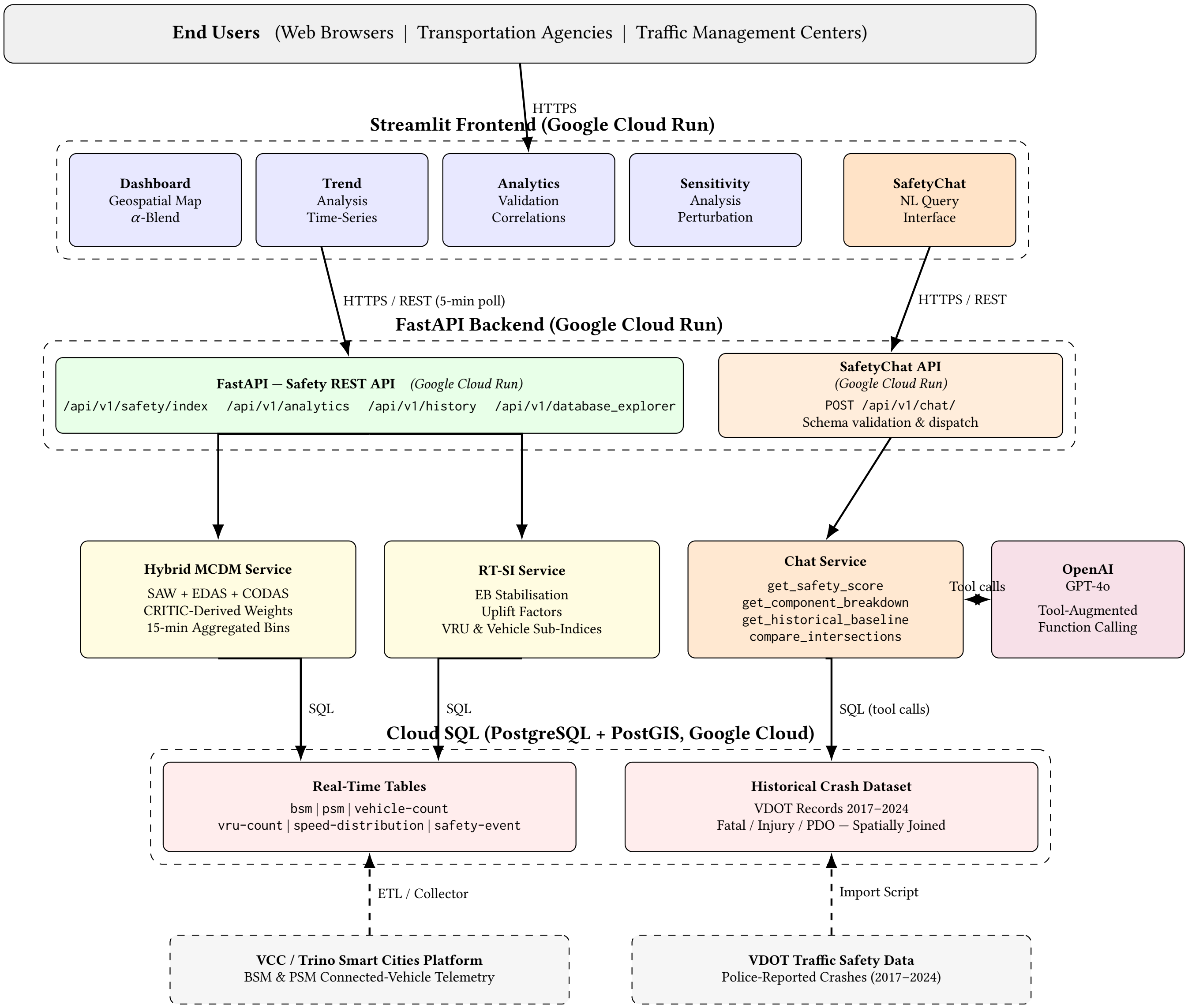


Figure 1: VTTSI cloud-native system architecture. The orange pathway (SafetyChat API, Chat Service, OpenAI GPT-4o) is the new contribution of this demo paper; the blue/green/yellow left pathway shows the existing safety-index pipeline.

where $C_{i,s}$ are crash counts by severity and E_i is vehicle exposure. The EB stabilized rate is:

$$\hat{r}^{\mathrm{EB}}_{i,t}(\lambda) = \frac{Y_{i,t} + \lambda\, r_0}{1 + \lambda}, \quad \lambda^\star = 10{,}000, \tag{2}$$

where r_0 is the global mean rate across all site-bins (2017–2024) and $\lambda^\star$ is chosen by minimizing Poisson log-loss on a 2025 hold-out set. Real-time uplift factors—bounded to $[0, 1]$—adjust this baseline for speed deficit (F^{spd}), speed variance (F^{var}), and VRU–vehicle conflict exposure (F^{conf}):

$$U_{i,t} = 1 + \beta_1 F^{\mathrm{spd}}_{i,t} + \beta_2 F^{\mathrm{var}}_{i,t} + \beta_3 F^{\mathrm{conf}}_{i,t}, \tag{3}$$

with $(\beta_1, \beta_2, \beta_3) = (0.3, 0.3, 0.4)$. VRU and vehicle sub-indices are combined with weights $(\omega_{\mathrm{vru}}, \omega_{\mathrm{veh}}) = (0.6, 0.4)$ and min-max scaled to $\mathrm{SI}^{\mathrm{RT}} \in [0, 100]$.

2.2 MCDM Safety Index

A CRITIC-weighted multi-criteria pipeline [1] constructs a decision matrix over five real-time criteria (vehicle count, VRU count, average speed, speed variance, incident count) for each intersection–time-bin pair. CRITIC criterion weights are derived from normalized standard deviations and inter-criterion Pearson correlations, recomputed every 24 hours. Three MCDM scoring methods—SAW, EDAS, and CODAS—are each computed and scaled to $[0, 100]$; a second-pass CRITIC aggregation yields the hybrid index $\mathrm{SI}^{\mathrm{MCDM}} \in [0, 100]$

Table 1: VTTSI-Chat vs. Related Systems

System	Hybrid Index	Real-Time Telemetry	LLM Interface	Causal Narrative
Google / Apple Maps	✗	✓	✗	✗
Waycare / ATSPM	✗	✓	✗	✗
TrafficGPT [12]	✗	✓	✓	✗
LLMLight [5]	✗	✓	✓	✗
GPT-Driver [6]	✗	✓	✓	✗
VTTSI-Chat (ours)	✓	✓	✓	✓

(higher = higher risk). Validation over 18 intersections shows that the three methods achieve pairwise Pearson correlations $r \approx 0.78$–0.99, and RT–SI has a mean absolute deviation of ≈ 0.09 under $\pm 25\%$ joint parameter perturbation [9].

2.3 Blended Final Index

$$\mathrm{SI}_{i,t}^{\mathrm{Final}} = \alpha\, \mathrm{SI}_{i,t}^{\mathrm{RT}} + (1-\alpha)\, \mathrm{SI}_{i,t}^{\mathrm{MCDM}}, \quad \alpha = 0.7 \text{ (default)}. \tag{4}$$

Scores map to risk levels: 0–40 = low, 41–70 = moderate, 71–100 = high.

2.4 Geospatial Foundation and Dashboard

Each of the 18 intersections is registered with a centroid (lat_i, lon_i) and an axis-aligned bounding box B_i ($\delta \approx 0.001°$, $\approx$111 m) in PostGIS. The one-time spatial join materialises per-site counts $C_{i,s}$ (Eq. (1)) as a static lookup table. The Streamlit Dashboard polls `/api/v1/safety/index` every five minutes and recolors each `folium.CircleMarker`—green (SI < 60), amber (60–75), red (> 75)—with radius proportional to traffic volume, giving operators a dual spatial encoding of risk and exposure. An interactive α-blending slider lets operators shift the final index between the historical-risk (RT–SI) and operational-anomaly (MCDM) poles in real time (Figure 2a).

3 SafetyChat: LLM-Powered Explanation Module

VTTSI already answers *what* the score is; SafetyChat answers *why* and *what to do*. GPT-4o is equipped with typed function calls [8] that map to VTTSI REST endpoints, grounding every claim in live geospatial data. An agentic loop (up to six iterations) enables chained calls—e.g., ranking all intersections by spatial risk and then drilling into the top site's component breakdown. The six tools are: (1) `get_safety_score` — RT–SI, MCDM, and blended scores with uplift factors; optional `target_time` for historical queries; (2) `get_component_breakdown` — CRITIC-weighted criteria for a 15-min bin; (3) `get_historical_baseline` — EB prior and 7-year crash severity breakdown; (4) `compare_intersections` — ranks all sites by any index component; (5) `get_trend_data` — 7–30 day MCDM trend with sampled snapshots; (6) `run_sql_query` — read-only ad-hoc SQL over the PostGIS database, write-guarded.

Table 1 positions VTTSI-Chat against related systems.

Table 2: Summary of Demonstration Use Cases

Use Case	Key Tool Calls	User
Morning briefing	compare + breakdown	TMC Operator
Causal explanation	score(t) + breakdown(t)	Planner
Responder routing	score × 2	Dispatcher
Trend engagement	get_trend_data(30d)	Official/Public

4 Demonstration

4.1 Demo Setup

The live demo runs against the production VTTSI deployment monitoring 18 intersections in Falls Church, VA. Conference attendees interact directly with the SafetyChat interface via a browser, typing natural language queries and observing LLM tool-call traces in real time alongside the companion Streamlit geospatial dashboard. Four canonical use cases guide the session.

4.2 UC1: TMC Operator Morning Briefing

Query: "Give me a morning safety briefing for all intersections." SafetyChat chains `compare_intersections` and `get_component_breakdown` for the two highest-ranked sites, delivering a situational-awareness narrative identifying peak-risk windows and dominant criteria (e.g., speed variance, VRU counts) in under five seconds. Where the Streamlit dashboard requires navigating multiple spatial charts, SafetyChat synthesizes multi-intersection data into a single paragraph.

4.3 UC2: Causal Safety Explanation for Planners

Query: "Why did E. Broad–N. Washington score above 70 yesterday at 5 PM?" SafetyChat queries `get_safety_score` and `get_component_breakdown` at the historical time-bin, identifies criteria in the 90th percentile, and traces the elevated score to concurrent peak vehicle volume and incident activity.

4.4 UC3: Emergency Responder Routing

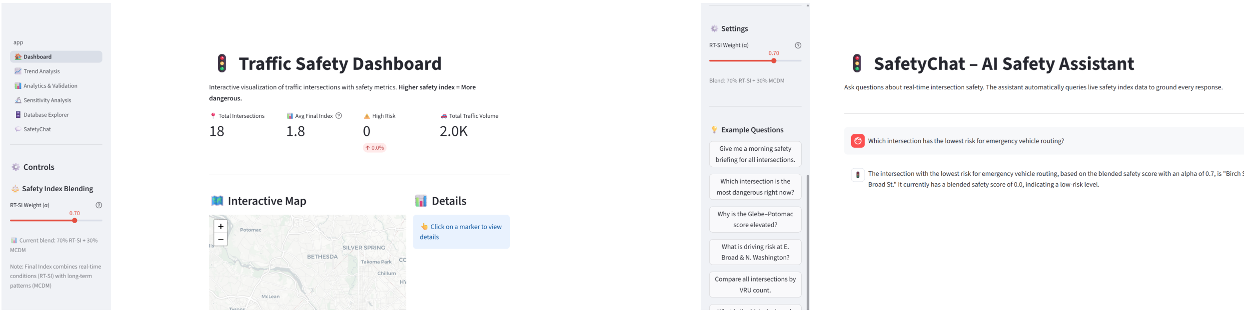
Query: "Which of Glebe–Potomac or Birch–W. Broad has lower current risk?" SafetyChat compares live RT–SI, MCDM, and blended indices for both intersections and recommends the lower-risk route with reasoning grounded in VRU counts and speed variance.

4.5 UC4: Public Stakeholder Trend Engagement

Query: "Is the Glebe–Potomac intersection getting safer over time?" SafetyChat calls `get_trend_data` (30-day window) and translates mean/max/min MCDM scores into plain-language risk levels with a trend direction indicator, making geospatial safety data accessible to non-technical stakeholders.

5 Related Work

Crash-based and hybrid safety indices. VTTSI's scoring builds on EB-stabilized crash models [7, 9], MCDM-based indices [1, 2], and surrogate safety indicators [4, 11]. Existing hybrid indices produce numerical outputs only, with no conversational or geospatial explanation layer.



(a) Dashboard: interactive Folium map with 18 monitored intersections (Falls Church, VA), real-time KPI cards (Avg Index 1.8, 0 High-Risk sites, 2.0 K traffic volume), and α-blending slider.

(b) SafetyChat (UC3): emergency routing query—GPT-4o calls `compare_intersections` *and returns Birch & W. Broad as the safest crossing (blended score 0.0) for immediate vehicle dispatch.*

Figure 2: Live VTTSI-Chat demonstration: (a) geospatial dashboard providing system-wide situational awareness; (b) SafetyChat natural-language response grounding an emergency routing recommendation in live blended safety indices.

LLMs in transportation. TrafficGPT [12], LLMLight [5], and GPT-Driver/DriveVLM [6, 13] address flow analysis, signal control, and AV planning respectively—none provides a safety-index explanation interface for infrastructure operators or operates over a PostGIS-backed spatio-temporal dataset.

Positioning. VTTSI-Chat is differentiated by combining: (i) a *validated hybrid geospatial safety index* fusing long-term crash history with 15-minute connected-vehicle telemetry; (ii) a *tool-augmented LLM* that accesses live PostGIS-backed API data; and (iii) *causal spatial narratives* that trace score components back to raw geospatial signals. No existing system offers all three, placing VTTSI-Chat squarely in the emerging area of agentic GeoAI and natural language interfaces for spatial data.

6 Conclusion

VTTSI-Chat demonstrates a working agentic GeoAI system that delivers causally grounded, natural language safety narratives over live spatio-temporal intersection data. By grounding all LLM outputs in six typed API calls to a PostGIS-backed safety pipeline, SafetyChat avoids hallucination while retaining the flexibility of conversational interaction over spatial data. The spatially-joined crash baseline, bounding-box intersection registry, and live 15-minute telemetry together form the *foundational geospatial layer* for future work integrating the Agentic Traffic Safety Knowledge Graph (TS–KG) [3]—enabling the LLM to reason over intersection geometry, conflict topology, and causal trajectory motifs, advancing the vision of trustworthy, explainable GeoAI for public safety.

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References

[1] A. Amraji. 2025. Combined Safety Index Using CRITIC Weighting. *Journal of Transportation Research* 15, 2 (2025), 200–210.

[2] Morteza AsadAmraji, Azarakhsh Salem, and Shila Shirinbayan. 2025. A Combined Index of Proactive and Reactive Data for Rating the Safety of Road Sections. *Iranian Journal of Science and Technology – Transactions of Civil Engineering* 49, 1 (2025), 985–995. doi:10.1007/s40996-024-01552-0

[3] Jason Cusati. 2025. *Agentic Knowledge Graphs for Traffic Safety.* Technical Report. Virginia Tech. Unpublished manuscript.

[4] Douglas Gettman and Larry Head. 2003. *Surrogate Safety Assessment Model and Validation: Final Report.* Technical Report. FHWA.

[5] Siqi Lai, Zhao Chen, et al. 2024. LLMLight: Large Language Models as Traffic Signal Control Agents. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining.* doi:10.48550/arXiv.2312.16171

[6] Jiageng Mao, Yuxi Qian, Hang Zhao, and Yue Wang. 2023. GPT-Driver: Learning to Drive with GPT. In *Advances in Neural Information Processing Systems (NeurIPS) Workshop.* doi:10.48550/arXiv.2310.01415

[7] Alfonso Montella, Carmen Guida, Jlenia Mosca, Jaeyoung Lee, and Mohamed Abdel-Aty. 2020. Systemic Approach to Improve Safety of Urban Unsignalized Intersections: Development and Validation of a Safety Index. *Accident Analysis & Prevention* 141 (2020), 105523. doi:10.1016/j.aap.2020.105523

[8] OpenAI. 2024. Function Calling in the OpenAI API. OpenAI Platform Documentation. https://platform.openai.com/docs/guides/function-calling

[9] Bhagwant Persaud and Craig Lyon. 2007. Empirical Bayes Before–After Safety Studies: Lessons Learned from Two Decades of Experience and Future Directions. *Accident Analysis & Prevention* 39, 3 (2007), 546–555. doi:10.1016/j.aap.2006.09.009

[10] Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, et al. 2023. Toolformer: Language Models Can Teach Themselves to Use Tools. *Advances in Neural Information Processing Systems* 36 (2023). doi:10.48550/arXiv.2302.04761

[11] John Schultz. 2025. A Study on Surrogate Safety Indicators. In *Journal of Transportation Safety*, Vol. 12. 123–134.

[12] Ning Shi, Xu Zhao, and Junfeng Zheng. 2023. TrafficGPT: Viewing, Processing and Interacting with Traffic Foundation Models. In *Proceedings of the 31st ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems.* doi:10.48550/arXiv.2309.06719

[13] Xiaoyu Tian, Junru Gu, Bailin Li, et al. 2024. DriveVLM: The Convergence of Autonomous Driving and Large Vision-Language Models. In *Conference on Robot Learning (CoRL).* doi:10.48550/arXiv.2402.12289